

D3: Scalar $q(\tilde{\beta})$ Update Under $\eta = \mathbf{YF}\tilde{\beta}$

Cluster B (Cox-on-YF Reformulation) — Derivation D3

Cross-reference: `derivations/Modular_UpdateDerivations/ q( $\beta$ )` section

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1 Context and Goal

In the original model, the Cox linear predictor for factor k uses the latent loading: $\eta_{\text{no-}k,i} + l_{ik} \beta_k$. The $q(\beta_k)$ update therefore has EBNM precision involving $\mathbb{E}[l_{ik}^2]$ (the second moment of the posterior, which includes shrinkage uncertainty).

Under Cluster B, the linear predictor becomes $\eta_i = \sum_k (z_F)_{ik} \tilde{\beta}_k$ where $(z_F)_{ik} = \sum_j y_{ij} \mathbb{E}[f_{jk}]$ is an *observed* projection score. Because $\mathbf{Z}_F$ is treated as fixed (held at $\mathbf{Y}\mathbb{E}[\mathbf{F}]$) per outer CAVI iteration, there is **no second-moment correction needed for the predictor** — $\mathbf{Z}_F$ has zero posterior variance from $q(\tilde{\beta}_k)$'s perspective.

Notation: the projection score $\mathbf{Z}_F$ (clarification added 2026-10-01). $\mathbf{Z}_F = \mathbf{Y}\mathbb{E}[\mathbf{F}]$ is the $n \times K$ matrix with entries $(z_F)_{ik} = \sum_j y_{ij} \mathbb{E}[f_{jk}]$: patient i 's expression profile projected onto the posterior-mean loading vector of program k . The subscript F records that the score is built from the loadings $\mathbf{F}$, as opposed to the latent score $\mathbf{L}$. The two have different roles: $\mathbf{L}$ appears only in the reconstruction $\mathbf{Y} \approx \mathbf{L}\mathbf{F}^\top$, and $\mathbf{Z}_F$ appears only in the Cox linear predictor. Because $\mathbf{Z}_F$ is a function of observed $\mathbf{Y}$, it can be computed for a new patient without estimating that patient's $\mathbf{L}$.

Normalization in the implementation (`code/fit_cox_on_yf.R`). The columns of $\mathbb{E}[\mathbf{F}]$ are scaled to unit Euclidean norm before projecting:

$$\hat{\mathbf{Z}}_F = \mathbf{Y}\mathbb{E}[\mathbf{F}]\mathbf{D}^{-1}, \quad \mathbf{D} = \text{diag}(\|\mathbb{E}[\mathbf{f}_1]\|, \dots, \|\mathbb{E}[\mathbf{f}_K]\|),$$

and $\eta_i = \sum_k (\hat{z}_F)_{ik} \tilde{\beta}_k$. This only changes the units of $\tilde{\beta}$: $\hat{\mathbf{Z}}_F \tilde{\beta} = \mathbf{Y}\mathbb{E}[\mathbf{F}](\mathbf{D}^{-1}\tilde{\beta})$, so $\tilde{\beta}_k$ equals $\|\mathbb{E}[\mathbf{f}_k]\|$ times the coefficient on the unnormalized score. It was introduced because $\|(\mathbf{z}_F)_{\cdot k}\|$ grows with p . On that scale the spike-and-slab prior on $\tilde{\beta}$ drove $\mathbb{E}[\tilde{\beta}]$ to zero. Prediction applies the stored norms (`EF_norms`). Everything in this document holds with $(z_F)_{ik}$ read as $(\hat{z}_F)_{ik}$. The multimodal extension (`derivations/multimodal_YFB/`) uses the unnormalized $\mathbf{Z} = \sum_m \mathbf{Y}_m \mathbb{E}[\mathbf{F}_m]$ and fixes scale by rescaling each factor instead.

Goal of D3. Derive the EBNM inputs $A_{\tilde{\beta},k}$ and $B_{\tilde{\beta},k}$, show that α cancels in the pseudo-observation $x_{\tilde{\beta},k} = B/A$, and confirm the connection to the original $q(\beta)$ derivation.

2 Derivation

2.1 Isolating Factor k in the Survival ELBO

The Taylor-linearised Cox surrogate (evaluated at the current η) is

$$\mathbb{E}_q[\log p(\mathbf{t}, \boldsymbol{\delta})] \approx \sum_i \left[u_i \eta_i - \frac{1}{2} w_i \eta_i^2 \right] + \text{const.}$$

Write $\eta_i = (z_F)_{ik} \tilde{\beta}_k + z_{\text{no-}k,i}$, where

$$z_{\text{no-}k,i} = z_i - \sum_{k' \neq k} (z_F)_{ik'} \mathbb{E}[\tilde{\beta}_{k'}]$$

is the partial working response with factor k removed. Expanding η_i^2 :

$$\begin{aligned} \sum_i \left[u_i \eta_i - \frac{1}{2} w_i \eta_i^2 \right] \Big|_{\tilde{\beta}_k} &= \sum_i \left[\underbrace{u_i (z_F)_{ik} \tilde{\beta}_k}_{\propto B} - \frac{1}{2} \underbrace{w_i (z_F)_{ik}^2 \tilde{\beta}_k^2}_{\propto A} + w_i (z_F)_{ik} z_{\text{no-}k,i} \tilde{\beta}_k \right] + \text{const} \\ &= -\frac{1}{2} \tilde{\beta}_k^2 \underbrace{\sum_i w_i (z_F)_{ik}^2}_{A_{\tilde{\beta},k}/\alpha} + \tilde{\beta}_k \underbrace{\sum_i w_i z_{\text{no-}k,i} (z_F)_{ik}}_{B_{\tilde{\beta},k}/\alpha} + \text{const}, \end{aligned} \quad (1)$$

where the working response identity $z_i = \eta_i + u_i/w_i$ gives $u_i = w_i(z_i - \eta_i)$, so $u_i (z_F)_{ik} + w_i (z_F)_{ik} z_{\text{no-}k,i}$ combines to $w_i (z_F)_{ik} z_{\text{no-}k,i}$ after absorbing the η_i term.

2.2 EBNM Precision and Pseudo-Observation

Including the mixing weight α , the factor- k ELBO terms for $q(\tilde{\beta}_k)$ are proportional to:

$$A_{\tilde{\beta},k} = \alpha \sum_i w_i (z_F)_{ik}^2, \quad (2)$$

$$B_{\tilde{\beta},k} = \alpha \sum_i w_i z_{\text{no-}k,i} (z_F)_{ik}. \quad (3)$$

The EBNM pseudo-observation and scale are:

$$x_{\tilde{\beta},k} = \frac{B_{\tilde{\beta},k}}{A_{\tilde{\beta},k}} = \frac{\sum_i w_i z_{\text{no-}k,i} (z_F)_{ik}}{\sum_i w_i (z_F)_{ik}^2}, \quad s_{\tilde{\beta},k} = \frac{1}{\sqrt{A_{\tilde{\beta},k}}} = \frac{1}{\sqrt{\alpha \sum_i w_i (z_F)_{ik}^2}}. \quad (4)$$

α cancellation in $x_{\tilde{\beta},k}$

α appears in both $A_{\tilde{\beta},k}$ and $B_{\tilde{\beta},k}$ with the same factor, so it cancels in the ratio:

$$x_{\tilde{\beta},k} = \frac{\alpha \sum_i w_i z_{\text{no-}k,i} (z_F)_{ik}}{\alpha \sum_i w_i (z_F)_{ik}^2} = \frac{\sum_i w_i z_{\text{no-}k,i} (z_F)_{ik}}{\sum_i w_i (z_F)_{ik}^2}.$$

The point estimate (posterior mode) is α -invariant. α only controls the scale $s_{\tilde{\beta},k} = (\alpha \sum w_i (z_F)_{ik}^2)^{-1/2}$, which governs the amount of shrinkage toward zero. This is identical to the α -cancellation in the original $q(\beta)$ derivation, with $(z_F)_{ik}$ replacing l_{ik} .

3 Key Differences from Original $q(\beta)$

Quantity	Original ($\eta = \mathbf{L}\beta$)	Cluster B ($\eta = \mathbf{Z}_F \tilde{\beta}$)
Predictor	$l_{ik} = \mathbb{E}_q[l_{ik}]$ (latent, shrunk)	$(z_F)_{ik} = \sum_j y_{ij} \mathbb{E}[f_{jk}]$ (observed)
A_k	$\alpha \sum_i w_i \mathbb{E}[l_{ik}^2]$	$\alpha \sum_i w_i (z_F)_{ik}^2$
B_k	$\alpha \sum_i w_i z_{\text{no-}k,i} \mathbb{E}[l_{ik}]$	$\alpha \sum_i w_i z_{\text{no-}k,i} (z_F)_{ik}$
2nd moment	$\mathbb{E}[l_{ik}^2] = \text{Var}_q(l_{ik}) + \mathbb{E}[l_{ik}]^2$ needed	$(z_F)_{ik}^2$ — exact, no second moment

Why no second moment for $(z_F)_{ik}$: The original A_k used $\mathbb{E}[l_{ik}^2]$ because l_{ik} is a latent quantity with posterior uncertainty. The projection score $(z_F)_{ik} = \sum_j y_{ij} \mathbb{E}[f_{jk}]$ treats $\mathbb{E}[\mathbf{f}_k]$ as a

fixed constant (Gauss-Seidel contract) and y_{ij} as observed data — there is no posterior variance to account for from $q(\tilde{\beta}_k)$'s perspective. This simplifies the computation and removes the `EL2_k` (or `EL2` matrix) argument from `update_beta_k()`.

Compute z_no_k : The function `compute_z_no_k` changes from

$$z_{no-k,i} = z_i - (\mathbf{L}\boldsymbol{\beta})_i + l_{ik}\beta_k$$

to

$$z_{no-k,i} = z_i - (\mathbf{Z}_F\tilde{\boldsymbol{\beta}})_i + (z_F)_{ik}\tilde{\beta}_k.$$

Same algebraic structure; $\mathbf{Z}_F$ replaces $\mathbf{L}$ and $\tilde{\boldsymbol{\beta}}$ replaces $\boldsymbol{\beta}$.

4 EBNM Call

EBNM call for $q(\tilde{\beta}_k)$

Inputs (scalars):

$$x_{\tilde{\beta},k} = \frac{\sum_i w_i z_{no-k,i} (z_F)_{ik}}{\sum_i w_i (z_F)_{ik}^2}, \quad s_{\tilde{\beta},k} = \frac{1}{\sqrt{\alpha \sum_i w_i (z_F)_{ik}^2}}.$$

Prior family: point-normal (or as configured in `prior_beta`)

Outputs: $\mathbb{E}[\tilde{\beta}_k]$, $\mathbb{E}[\tilde{\beta}_k^2]$, `ebnm_log_likelihood`

5 Code Implications

`code/update_beta.R` — `compute_z_no_k()`

- Old: `compute_z_no_k(z, EL, EBeta, k)`
- New: `compute_z_no_k(z, ZF, EBeta, k)`
- Body: `z - (ZF %*% EBeta) + ZF[, k] * EBeta[k]`

`code/update_beta.R` — `update_beta_k()`

- **Remove argument:** `EL2_k` (no second moment needed)
- **Rename argument:** `EL_k` $\rightarrow$ `ZF_k` (n-vector of projection scores)
- **New signature:** `update_beta_k(w, z_no_k, ZF_k, prior_family, alpha, A_floor)`
- **New computation:**

```
A_k <- max(alpha * sum(w * ZF_k^2), A_floor) # ZF_k^2 exact (no second moment)
B_k <- alpha * sum(w * z_no_k * ZF_k)
```

`code/update_beta.R` — `update_beta_all()`

- Replace EL, EL2 with ZF ($n \times K$ matrix) in signature
- Remove all EL2 threading

6 Verification Checklist

D3 Verification

1. $A_{\tilde{\beta},k} = \alpha \sum_i w_i (z_F)_{ik}^2$ — matches (2). ✓
2. $B_{\tilde{\beta},k} = \alpha \sum_i w_i z_{\text{no-}k,i} (z_F)_{ik}$ — matches (3). ✓
3. α cancels in $x_{\tilde{\beta},k}$ — pseudo-observation is α -invariant. ✓
4. No EL2_k / second moment needed for the predictor. ✓
5. `compute_z_no_k` uses ZF instead of EL. (to implement)
6. `update_beta_k` signature reduced: EL2_k removed. (to implement)
7. All β tests pass with ZF predictor. (to implement)